

Jonathan Hawla

PhD student in Cosmology

CEA Paris-Saclay — IRFU / DPhP · DESI Collaboration · Université Paris-Saclay (ED 576 PHENIICS)

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Research Interests

Observational and computational cosmology: large-scale structure, Bayesian field-level inference with differentiable simulators, galaxy clustering $\times$ CMB lensing cross-correlations, primordial non-Gaussianity (f_{NL}), GPU-accelerated inference.

Education

2025 – 2028 (exp. Sep. 2028)	PhD in Cosmology — CEA Paris-Saclay, IRFU/DPhP, Saclay, France Université Paris-Saclay, Doctoral School ED 576 PHENIICS. Fully funded (<i>contrat doctoral</i> , 36 months). <i>Bayesian field-level inference with differentiable simulators for the joint analysis of DESI galaxy clustering and CMB lensing.</i> Supervisors: Étienne Burtin (director) & Arnaud De Mattia (co-supervisor).
2024 – 2025	M.Sc. (M2) Astronomy, Astrophysics & Space Engineering Observatoire de Paris – PSL, Paris–Meudon, France. Graduated with honours (<i>mention bien</i>).
2023 – 2024	M.Sc. (M1) Universe Sciences & Space Technologies Observatoire de Paris – PSL, Paris–Meudon, France. Graduated with honours (<i>mention bien</i>).
2020 – 2023	B.Sc. Physics Université de Franche-Comté, UFR Sciences et Techniques, Besançon, France. Graduated with honours (<i>mention bien</i>).

Research Experience

Oct. 2025 – present	Doctoral research — CEA Paris-Saclay (IRFU/DPhP), Saclay, France - Developed a fully differentiable, curved-sky forward model with a light cone (Lagrangian perturbation theory, galaxy bias, redshift-space distortions, Born-approximation lensing projector), in Python/JAX on GPUs at NERSC. - Extended the field-level inference framework of H. Simon et al. (2025) to CMB lensing, building on JaxPM (W. Kabalan); posterior sampling with a Microcanonical Langevin Monte Carlo (MCLMC) sampler via <code>numpyro</code>. - Identified and solved the <i>finite box</i> problem by analytically marginalising the out-of-box lensing contribution as a structured additive variance; implemented a pixel-exact likelihood enabling unbiased inference at small sky fractions. - Validated the pipeline at realistic noise levels (DESI LRG shot noise, ACT DR6 reconstruction noise) on synthetic closure data and on ABACUSSUMMIT N-body simulations, quantifying the gain brought by CMB lensing on σ_8, galaxy bias and local primordial non-Gaussianity f_{NL}. - Implemented survey mask and footprint handling, enabling inference on realistic partial-sky geometries. - Ongoing: model-misspecification studies and application to real data (DESI LRG $\times$ Planck PR4 / ACT DR6).
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2025 (3 months)	Research internship — CEA Paris-Saclay (DPhP), Saclay, France Supervisors: A. De Mattia & É. Burtin. Combined analysis of DESI 3D galaxy clustering and CMB lensing convergence maps from Planck and ACT.
2024 (2 months)	Research internship — LUTH, Observatoire de Paris, Meudon, France Supervisor: P.-S. Corasaniti. Dark-energy models with a rapid phase transition in the context of the Hubble tension; Planck CMB likelihood analysis with CLASS and MontePython to constrain the <i>kink</i> parametrisation.
2023 (2 months)	Research internship — FEMTO-ST, Besançon, France Supervisor: J. Gillot. Impact of optical power fluctuations on the stability of a laser stabilised by a cryogenic Fabry–Pérot cavity, in preparation for a power-servo loop.
2025 (1 week)	Observing run — Observatoire de Haute-Provence, France Spectroscopy of Be stars with the T152 telescope: observation planning, telescope operation, data reduction.
2025	Methodology project — Observatoire de Paris, Meudon, France Supervisor: Y. Rasera. Particle-Mesh N -body simulation of galaxy clustering implemented in Fortran; team project with Git-based version control.

Publications

No refereed publication yet. First-author paper in preparation on field-level inference for the joint analysis of DESI galaxy clustering and CMB lensing.

Talks

Jul. 2026	Adding CMB lensing to the galaxy clustering field-level analysis DESI Collaboration Meeting, Durham, UK — Primordial Non-Gaussianity working group, contributed talk.
Jun. 2026	Field-level inference for the joint analysis of galaxy clustering and CMB lensing GDR CoPhy annual meeting, Clermont-Ferrand, France — contributed talk.
Apr. 2026	Field-level inference for the joint analysis of galaxy clustering and CMB lensing Elbereth Conference, Institut d'Astrophysique de Paris, Paris, France — contributed talk.
Dec. 2025	Spotlight talk — DESI Collaboration Meeting, Tucson, AZ, USA.

Posters

Jun. 2026	Field-level inference for the joint analysis of galaxy clustering & CMB lensing PheniicsFest, annual meeting of the ED PHENIICS, Université Paris-Saclay, Orsay, France.
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Schools & Training

Aug. 2026	Rodolphe Clédassou (Euclid) Summer School — Cycle 2 (Theory) , 2 weeks, Pornichet, France.
Aug. 2025	Rodolphe Clédassou (Euclid) Summer School — Cycle 1 , 2 weeks, Les Karellis, France.
2025 – 2026	Doctoral training — Research ethics and scientific integrity (MOOC); prevention of gender-based and sexual violence and of discrimination. Université Paris-Saclay.

Collaboration Service

2025 – present **DESI Collaboration** — active member; contributor to the Primordial Non-Gaussianity working group. Three-night observing shift as remote Supporting Observer.

Teaching & Outreach

2024 – present **Private tutoring** — high-school mathematics (3 h/week) and piano (1 h/week).
2022 **Volunteer organiser** — SF2A annual meeting (French Society of Astronomy & Astrophysics), 1 week, Besançon, France.

Software & Projects

2025 – present **thesis-desi-cmb-fli**  — differentiable forward model and Bayesian inference pipeline for the joint field-level analysis of galaxy clustering and CMB lensing (Python/JAX, multi-GPU). Open source (MIT), with unit tests, continuous integration and documentation.
2022 – 2026 **CarbonScale** — designer and developer of an app to track and reduce personal CO₂ emissions. Winner of the *Pépîte Bourgogne-Franche-Comté* idea competition (2023).  [carbonscale.app](#)

Technical Skills

Languages	Python (NumPy, JAX, numpyro , Astropy, HEALPy), Fortran, Bash, L^AT_EX .
Cosmology	CLASS , MontePython, JaxPM, HEALPix , ABACUSSUMMIT simulations.
Methods	Bayesian inference, MCMC/HMC/MCLMC samplers, automatic differentiation, GPU computing (NERSC), <i>N</i> -body and LPT simulations.
Tools	Git, Linux/HPC environments, Keynote, Lightroom.

Languages

French — native speaker. **English** — fluent; TOEIC 965/990 (CEFR C1, 2023), EF SET 72/100 (C2, 2023); all talks given in English. **Lebanese Arabic** — fluent spoken (not written).

Interests

Piano (ten years at the conservatoire, band *Puls'art*) — first-aid volunteer, French Red Cross (PSE2) — Muay Thai — photography — chess — philosophy of science.

References

Étienne Burtin (etienne.burtin@cea.fr) and **Arnaud De Mattia** (arnaud.de-mattia@cea.fr), CEA Paris-Saclay, IRFU/DPhP.